Lab 3:

TASKS:

Task1: Simulate a discrete time continuous valued (DTCV) sinusoid of 1/50 cycles/sample with length of the signal to be 500 & choose the no. of significant digits for round-off and apply to the signal generated above. Compute the error signals and SQNR.

```
A = 1;

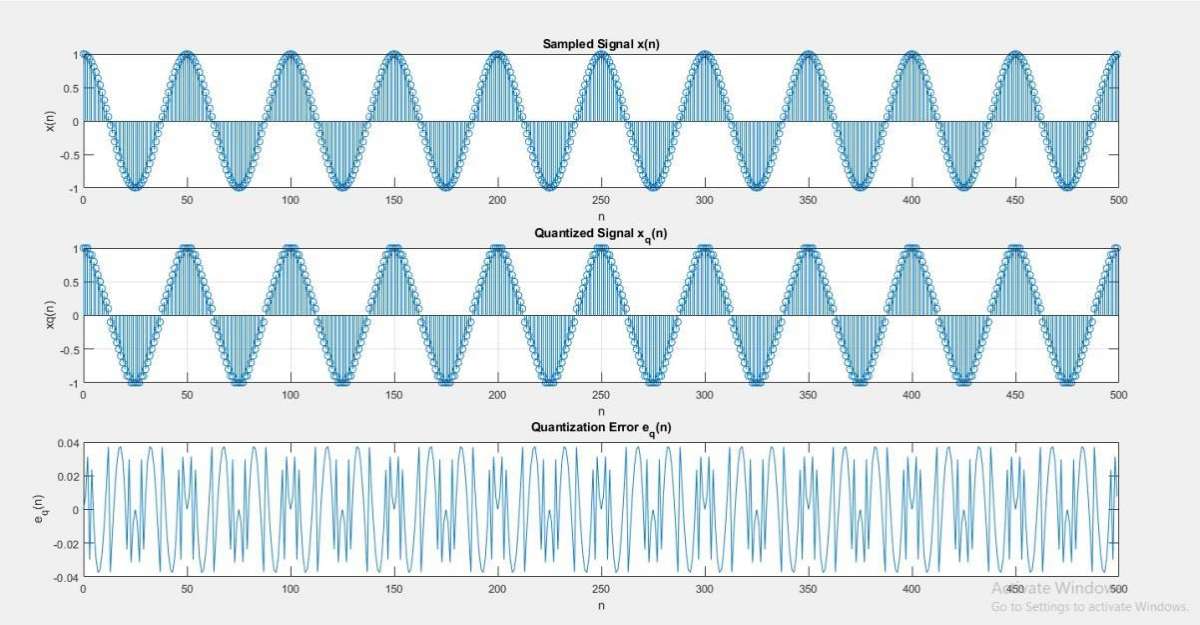
f0 =
1/50; N =
500;
n = 0:N-1;
x =
A*cos(2*pi*f0*n); d
= 1;
xq = round(x,d);
eq = xq - x;
Px = mean(x.^2);
Pq =
mean(eq.^2);
SQNR = 10*log10(Px/Pq);

subplot(3,1,1);
stem(n,x);
title('Sampled Signal x(n)');
xlabel('n'); ylabel('x(n)');

subplot(3,1,2);
stem(n,xq);
title('Quantized Signal
x_q(n)'); xlabel('n');
ylabel('xq(n)'); grid on;

subplot(3,1,3);
plot(n,eq);
title('Quantization Error e_q(n)');
xlabel('n'); ylabel('e_q(n)');
```

A = 1, f0 = 1/50, N = 500, and the signal was rounded off to d = 1 significant digit after the decimal point. This gives a computed SQNR of 28.72 dB. Output shown below:



SQNR: 28.7 dB

Task2: For fo = 1/50 and N = 200, write a program to quantize the signal x(n), using (a) Truncation, (b) Rounding to 64, 128 and 256 levels. In each case plot the signals x(n), xq(n) and eq(n) and also compute the corresponding SQNR.

```
N = 200;
f0 = 1/50;
n = 0:N-1;
x = cos(2*pi*f0*n);

L_values = [64 128 256];
Px = mean(x.^2);

for k = 1:3
L = L_values(k); xmin = -1; xmax = 1;
Delta = (xmax - xmin)/(L - 1);

% ROUNDING
xq_round = round((x - xmin)/Delta)*Delta + xmin; eq_round = xq_round - x;
Pq_round = mean(eq_round.^2); SQNR_round = 10*log10(Px/Pq_round);

% TRUNCATION
xq_trunc = floor((x - xmin)/Delta)*Delta + xmin; eq_trunc = xq_trunc - x;
```

```
Pq_trunc = mean(eq_trunc.^2); SQNR_trunc = 10*log10(Px/Pq_trunc);

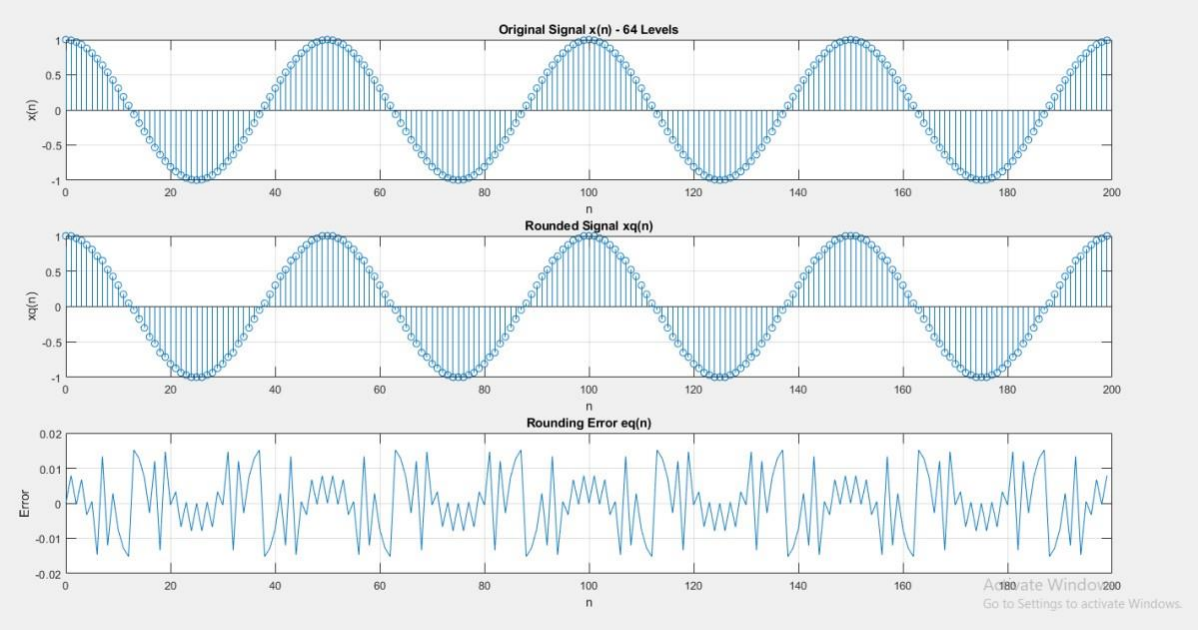
fprintf('\n   \n');
fprintf('Number of Levels = %d\n', L); fprintf('Delta = %.6f\n', Delta); fprintf('Rounding
SQNR = %.2f dB\n', SQNR_round);
fprintf('Truncation SQNR = %.2f dB\n', SQNR_trunc);

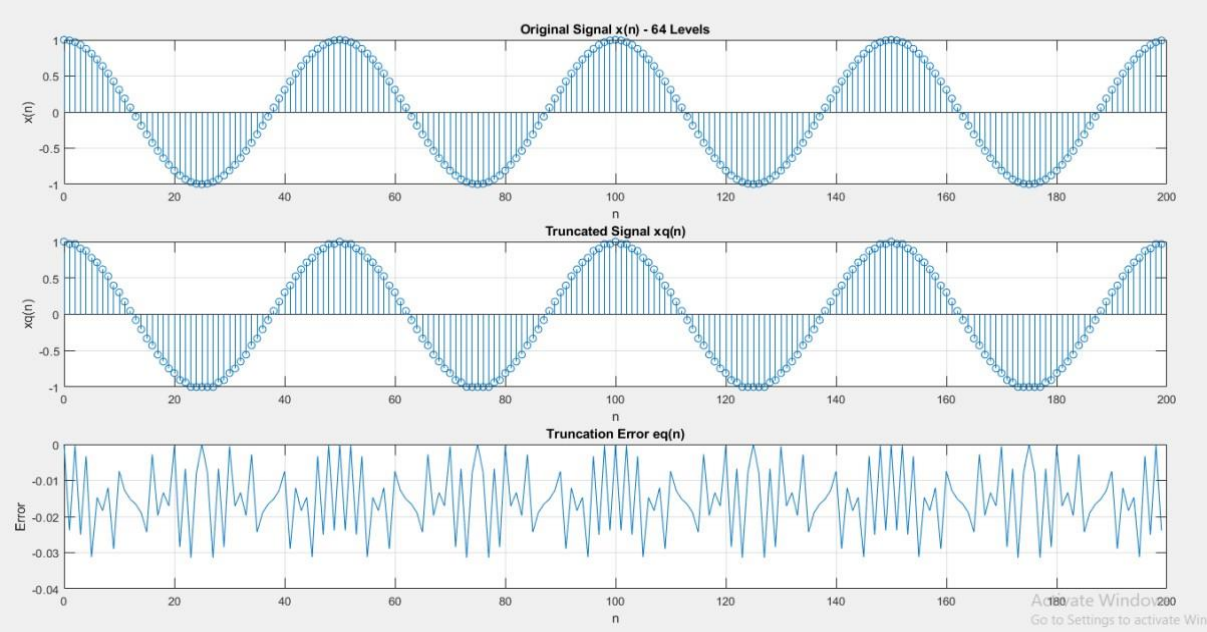
% PLOTS - ROUNDING
figure;
subplot(3,1,1); stem(n,x);
title(['Original Signal x(n) - ',num2str(L),' Levels']);
xlabel('n'); ylabel('x(n)'); subplot(3,1,2); stem(n,xq_round);
title('Rounded Signal xq(n)'); xlabel('n'); ylabel('xq(n)'); subplot(3,1,3);
plot(n,eq_round);
title('Rounding Error eq(n)'); xlabel('n'); ylabel('Error');

% PLOTS - TRUNCATION
figure;
subplot(3,1,1); stem(n,x);
title(['Original Signal x(n) - ',num2str(L),' Levels']);
xlabel('n'); ylabel('x(n)'); subplot(3,1,2); stem(n,xq_trunc);
title('Truncated Signal xq(n)'); xlabel('n'); ylabel('xq(n)'); subplot(3,1,3);
plot(n,eq_trunc);
title('Truncation Error eq(n)'); xlabel('n'); ylabel('Error');
end
```

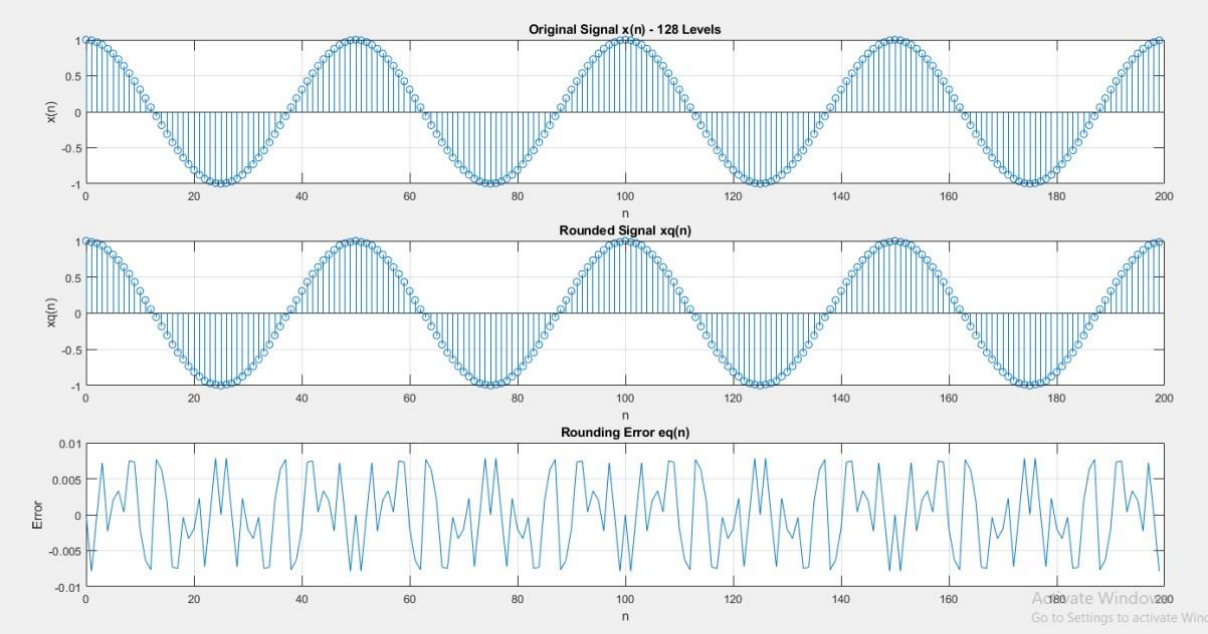
Levels (L)	Delta	SQNR - Rounding (dB)	SQNR - Truncation (dB)
64	0.031746	37.49	31.85
128	0.015748	42.57	38.02
256	0.007843	50.47	43.81

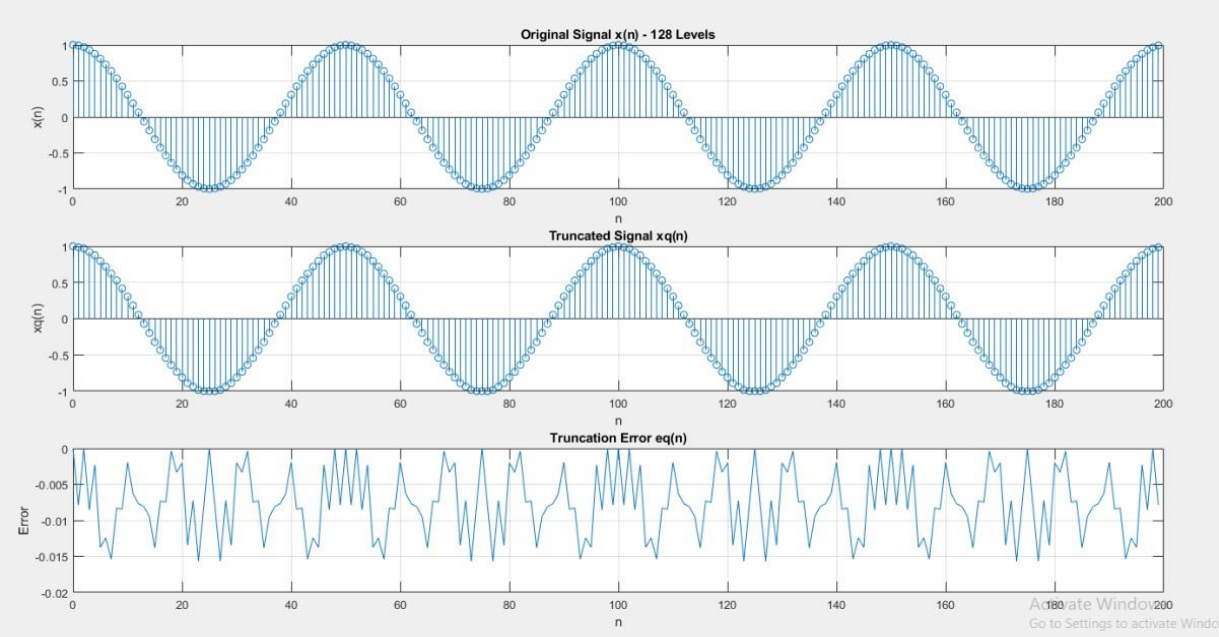
Results: 64 Quantization Levels





Results: 128 Quantization Levels





Results: 256 Quantization Levels

